

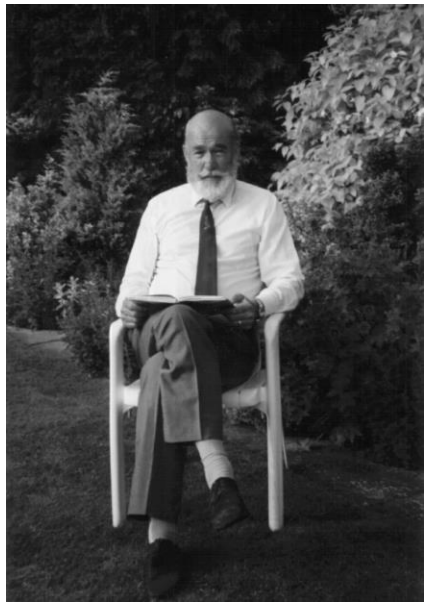
Lecture Series 3: ANOVA and GLMs CM2018

Adam S. Darwich

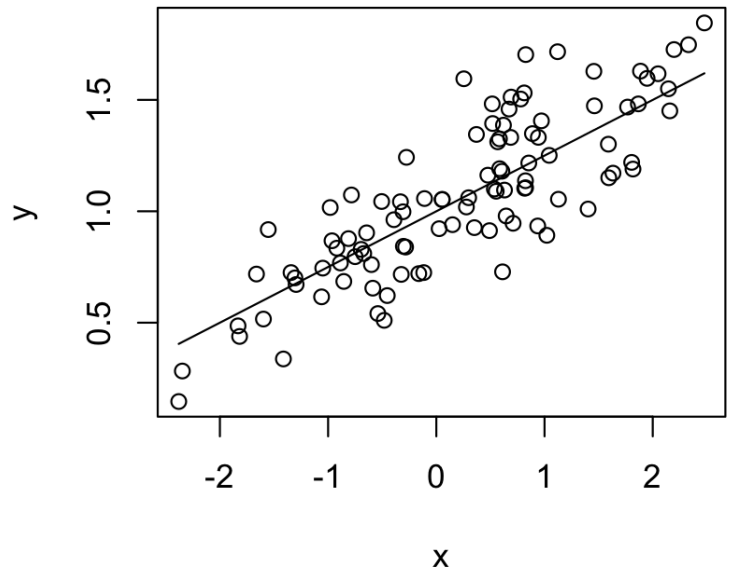
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L3-1 Generalised linear models

Generalised linear modelling







Generalized Linear Models

By J. A. NELDER and R. W. M. WEDDERBURN

Rothamsted Experimental Station, Harpenden, Herts

SUMMARY

The technique of iterative weighted linear regression can be used to obtain maximum likelihood estimates of the parameters with observations distributed according to some exponential family and systematic effects that can be made linear by a suitable transformation. A generalization of the analysis of variance is given for these models using log-likelihoods. These generalized linear models are illustrated by examples relating to four distributions; the Normal, Binomial (probit analysis, etc.), Poisson (contingency tables) and gamma (variance components).

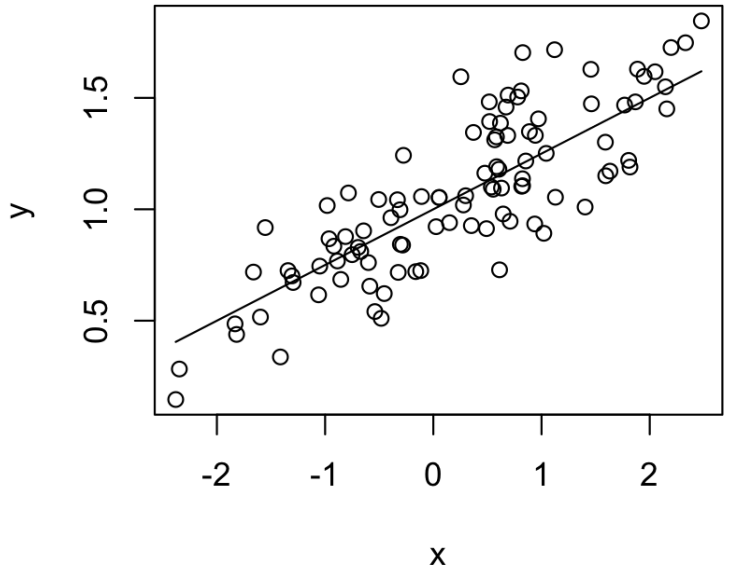
The implications of the approach in designing statistics courses are discussed.

Nelder & Wedderburn 1972; Senn 2003

Generalised linear modelling

$$y \sim \underbrace{Intercept + x \cdot Coefficient}_{systematic} + \underbrace{Error}_{random}$$

- A dependent variable z whose distribution with parameter θ is of the exponential family of distributions.
- A linear systematic component $Y = \sum \beta_i x_i$, η.
- A linking function $\theta = f(Y)$ connecting the parameter θ of the distribution of z with *Y*'s of the linear model.
- We estimate these models with maximum likelihood estimation.
- For an ordinary linear model z is normally distributed with a mean of θ and variance of $\sigma^2, \theta = Y$ with Normal errors.



Nelder & Wedderburn 1972

Link functions

Distribution		*Canonical Link:* $\theta = g(\mu)$	*Inverse Link:* $\mu = g^{-1}(\theta)$
Poisson		$\log(\mu)$	$\exp(\theta)$
Binomial	*logit link:*	$\log\left(\frac{\mu}{1-\mu}\right)$	$\frac{\exp(\theta)}{1+\exp(\theta)}$
	probit link:	$\Phi^{-1}(\mu)$	$\Phi(\theta)$
	cloglog link:	$\log(-\log(1-\mu))$	$1-\exp(-\exp(\theta))$
Normal		μ	θ
Gamma		$-\frac{1}{\mu}$	$-\frac{1}{\theta}$
Negative binomial		$\log(1-\mu)$	$1-\exp(\theta)$

Gill 2001

Generalised linear modelling

Dataset of outcomes:

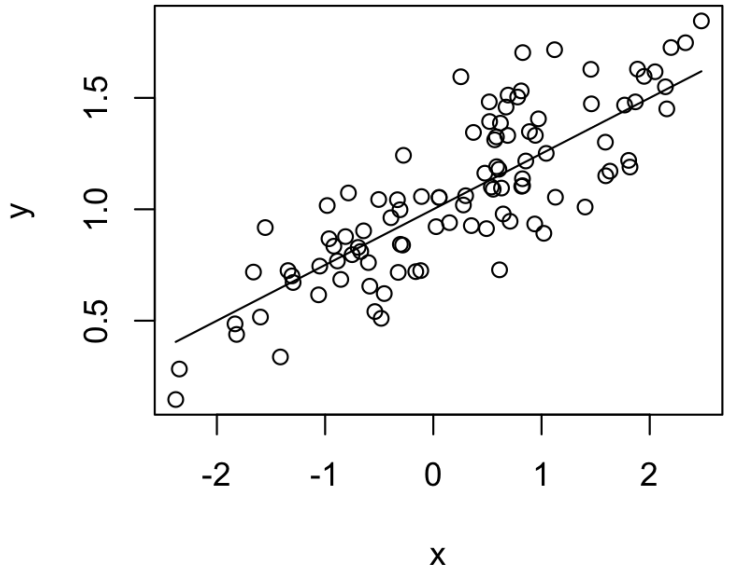
$y = (y_1, .., y_n)$

Data distribution:

$p(y|\hat{y})$

Link function:

$\hat{y} = E(y|X, \beta) = g^{-1}(X\beta)$



$\hat{y} = Intercept + x \cdot Coefficient$

Linear regression

Link function: $g(u) \equiv u$ where $u = \boldsymbol{X}\beta$

Normal distribution with standard deviation σ estimated from the data.

Gelman & Hill 2007

Generalised linear modelling

Dataset of outcomes:

$$y = (y_1, .., y_n)$$

A data distribution:

$$p(y|\hat{y})$$

A vector or matrix of predictors:

$$x = \begin{bmatrix} x_{11}x_{12} \cdots x_{1n} \\ x_{21}x_{22} \cdots x_{2n} \\ \vdots \quad \vdots \quad \ddots \quad \vdots \\ x_{m1}x_{m2} \cdots x_{mn} \end{bmatrix}$$

I. Link transformation using the canonical link function.
II. Maximum likelihood estimation of model parameters.

A vector of coefficients:

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_j \end{bmatrix}$$

I. Prediction of *y* on the original scale using the inverse link function of the model.
II. Estimation of the dispersion parameter ϕ on the normal scale.

A link function *g* yielding a vector of transformed data:

$$\hat{y} = E(y|X, \beta) = g^{-1}(\boldsymbol{X}\beta)$$

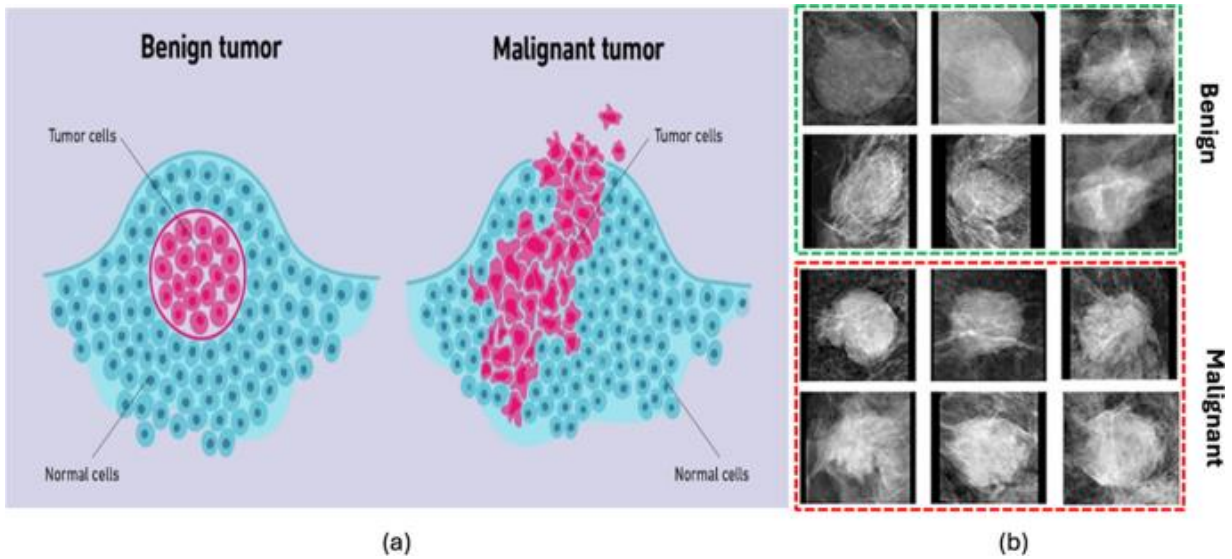
Nelder & Wedderburn 1972

Generalised linear modelling

- Logistic regression
- Poisson regression
- Ordered categorical regression
- Unordered categorical regression

L3-2 Logistic regression

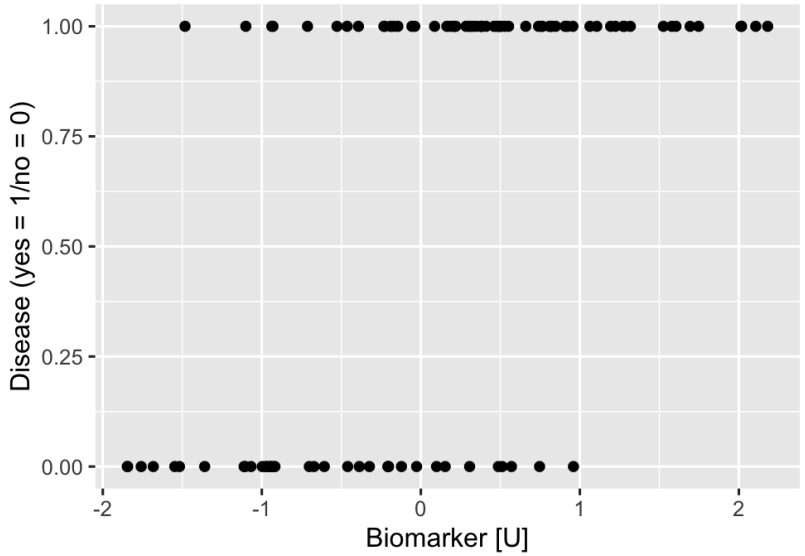
Binary outcomes



Islam *et al*. 2024

- Frequently present in medicine.
- Classification problems, presence of disease (yes/no) given a set of biomarkers.
- Clinical outcomes, success of failure of response to a given treatment and biomarkers.

Logistic regression



$$disease \sim \beta_0 + \beta_1 \cdot biomarker$$

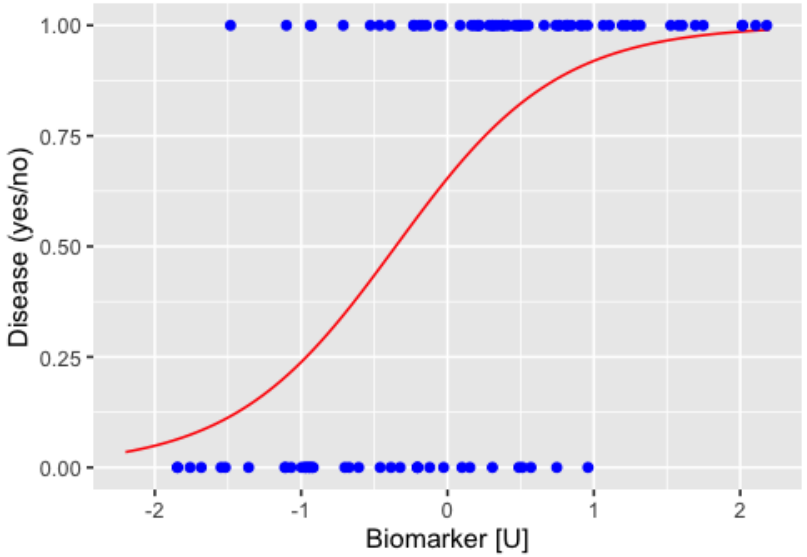
- Nonlinear transformation that bounds the output between 0 and 1.
- Produce an output that represent probabilities of success mapped to a random binary outcome.
- Using the generalised linear modelling approach, the regression model output is transformed using the *logit* function.
- We model the probability of $y = 1$,

$$P(y_i = 1) = logit^{-1}(X_i\beta)$$

Distribution		*Canonical Link:* $\theta = g(\mu)$	*Inverse Link:* $\mu = g^{-1}(\theta)$
Poisson		$\log(\mu)$	$\exp(\theta)$
Binomial	*logit link:*	$\log\left(\frac{\mu}{1-\mu}\right)$	$\frac{\exp(\theta)}{1+\exp(\theta)}$
	probit link:	$\Phi^{-1}(\mu)$	$\Phi(\theta)$
	cloglog link:	$\log(-\log(1-\mu))$	$1-\exp(-\exp(\theta))$
Normal		μ	θ
Gamma		$-\frac{1}{\mu}$	$-\frac{1}{\theta}$
Negative binomial		$\log(1-\mu)$	$1-\exp(\theta)$

Gelman & Hill 2007

An example of logistic regression in R




```
m1_binom <- glm(y ~ x1, family =
"binomial", data = binom_dat)
```

```
> summary(m1_binom)

Call:
glm(formula = y ~ x1, family = "binomial", data = binom_dat)

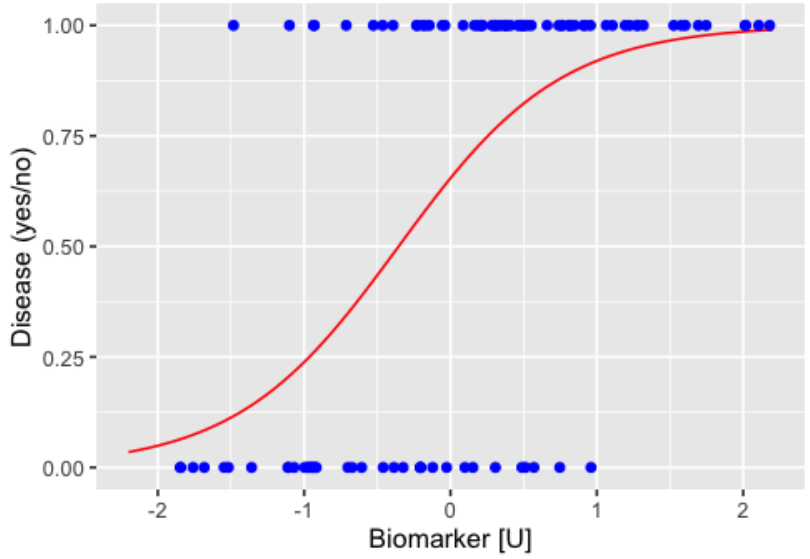
Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   0.6384     0.2624   2.433    0.015 *
x1            1.8008     0.3727   4.831 1.36e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 130.684  on 99  degrees of freedom
Residual deviance:  92.022  on 98  degrees of freedom
AIC: 96.022

Number of Fisher Scoring iterations: 5
```

Interpreting logistic regression models



- Coefficients are difficult to interpret because of nonlinearity

$$logit(0.1) = -2.20, \quad logit(0.2) = -1.39$$
$$logit(0.5) = 0, \quad logit(0.6) = 0.41$$

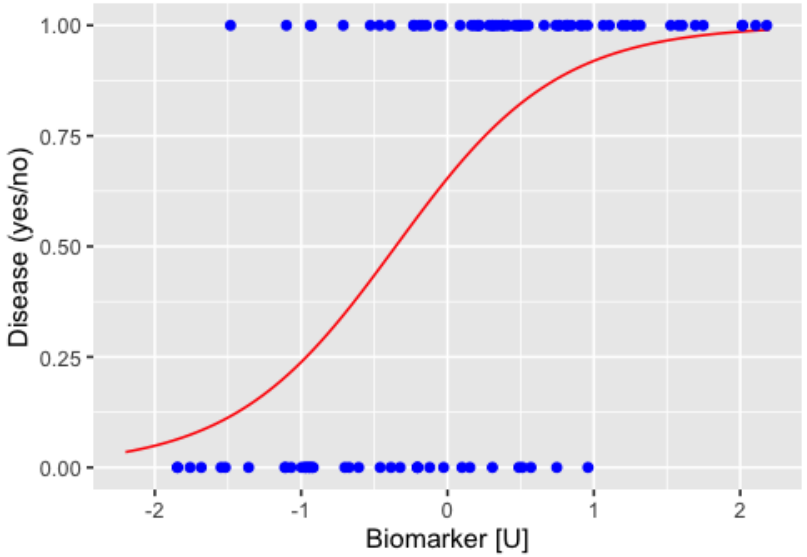
- We can test coefficient effects by calculating the probability, including the change at any part of the curve, *e.g.*:

```
p_i <- invlogit(coef(m1_binom)[1] + coef(m1_binom)[2]*…
mean(binom_dat$x1)), p_i = 0.70

p_d <- invlogit(coef(m1_binom)[1] + coef(m1_binom)[2]*1)) - …
invlogit(coef(m1_binom)[1] + coef(m1_binom)[2]*2)
p_d = 0.07
```

Gelman & Hill 2007

Interpreting logistic regression models



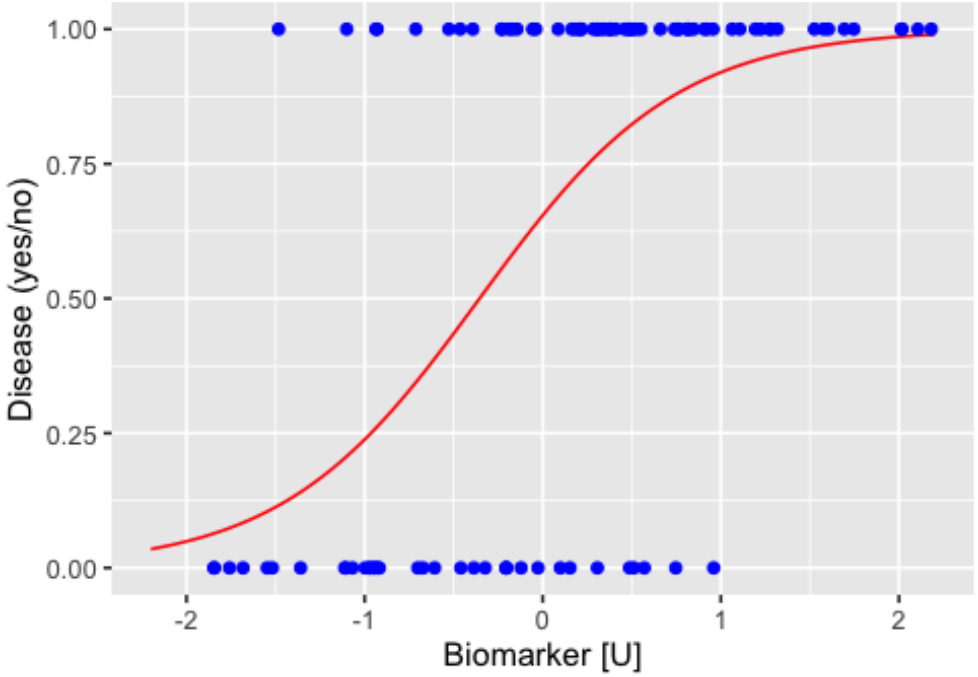
- **The divide-by-4 rule**
  The logistic curve is steepest at its center where $logit^{-1}(\beta_0 + \beta_1 x) = 0.5$
  The derivative of the logistic function (slope) is at this point:
  $\beta e^0/(1+e^0)^2 = \beta/4$ Thus, $\beta/4$ is the maximum difference in $P(y = 1)$ corresponding to a unit difference in x.

  $1.8 \,/\, 4 = 0.45$
  A difference in 1 in $Biomarker$ leads to a maximum positive difference in probability of 0.45 of $Disease$.

- **Odds ratios**
  If two outcomes have the probabilities p and $1 - p$, then $p/(1-p)$ is the odds.
  An odds of 1 is a probability of 0.5. The odds ratio is the ratio of two odds.
  For example, two different values of x: $^{p_1}/_{(1-p_1)}/^{p_2}/_{(1-p_2)}$

Gelman & Hill 2007

Prediction



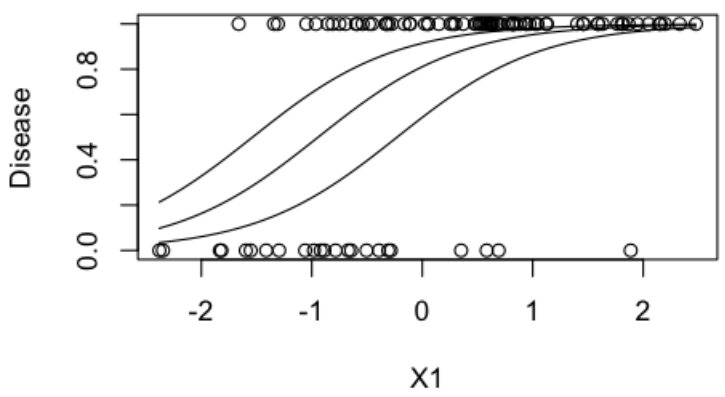

```r
# We can predict and plot model response
dat_new <- data.frame(x1=seq(from = -2.2, to = 2.2, length.out = 100))
pred <- predict(m1_binom, type="response", newdata = dat_new)

dat_pred <- data.frame(x1 = dat_new$x1, Predicted = pred, x2 = binom_dat$x1,
                       Observed = binom_dat$y)

ggplot() +
  geom_line(data = dat_pred, aes(x = x1, y = Predicted), color = "red") +
  geom_point(data = dat_pred, aes(x = x2, y= Observed), color = "blue") +
  xlab("Biomarker [U]") +
  ylab("Disease (yes/no)")
```

R

Prediction

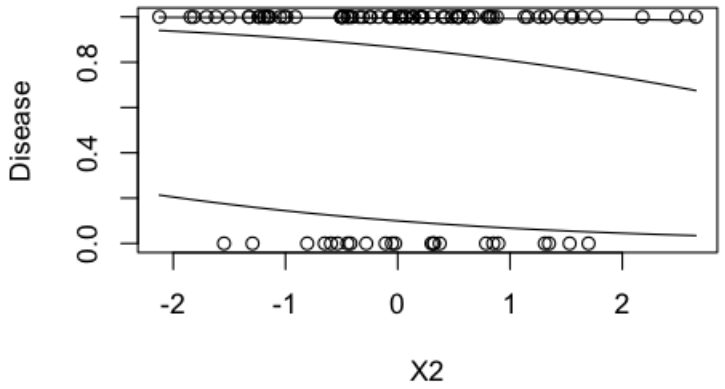


$$disease \sim \beta_0 + \beta_1 \cdot biomarker$$

```r
# Fitting the model
m3_binom <- glm(y ~ x1 + x2, family = "binomial", data = binom_dat)
summary(m3_binom)

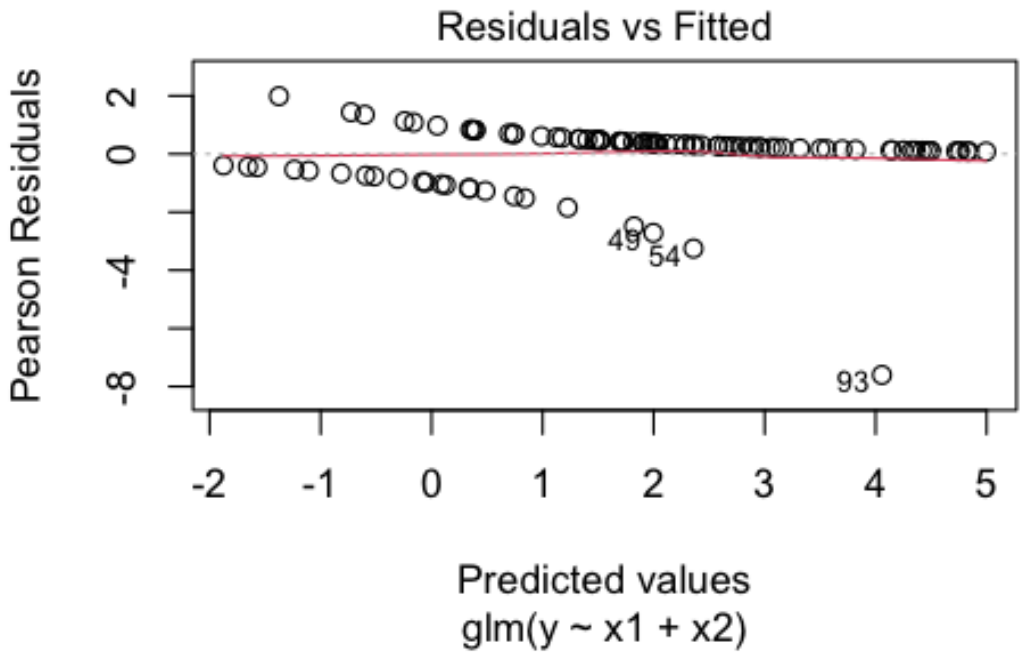
# Then plotting the calculated probability curves
plot(binom_dat$x1, binom_dat$y, xlim=c(min(binom_dat$x1),max(binom_dat$x1)), xlab = "X1",
ylab = "Disease")
curve(invlogit(cbind(1, x, min(binom_dat$x2)) %*% coef(m3_binom)), add=TRUE)
curve(invlogit(cbind(1, x, mean(binom_dat$x2)) %*% coef(m3_binom)), add=TRUE)
curve(invlogit(cbind(1, x, max(binom_dat$x2)) %*% coef(m3_binom)), add=TRUE)

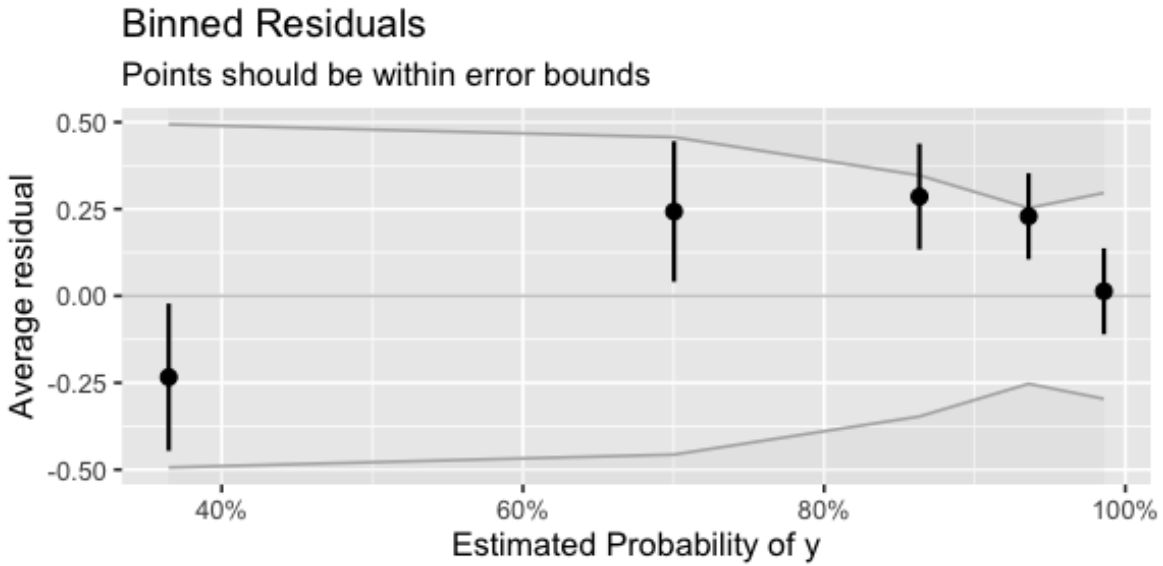
plot(binom_dat$x2, binom_dat$y, xlim=c(min(binom_dat$x2),max(binom_dat$x2)), xlab = "X2",
ylab = "Disease")
curve(invlogit(cbind(1, min(binom_dat$x1), x) %*% coef(m3_binom)), add=TRUE)
curve(invlogit(cbind(1, mean(binom_dat$x1), x) %*% coef(m3_binom)), add=TRUE)
curve(invlogit(cbind(1, max(binom_dat$x1), x) %*% coef(m3_binom)), add=TRUE)
```



$$disease \sim \beta_0 + \beta_1 \cdot biomarker_1 + \beta_2 biomarker_2$$

Residual errors






```
plot(m3_binom)
```

R

```
library(performance) # or library(DHARMa)
library(see)

m3_bres <- binned_residuals(
  m3_binom,
  term = NULL,
  n_bins = 5,
  show_dots = NULL,
  ci = 0.95,
  ci_type = "exact",
  residuals ="deviance",
  iterations = 1000,
  verbose = TRUE)

if (require("see")) {
  plot(m3_bres, show_dots = TRUE)
}
```

L3-3 Poisson regression

Count data

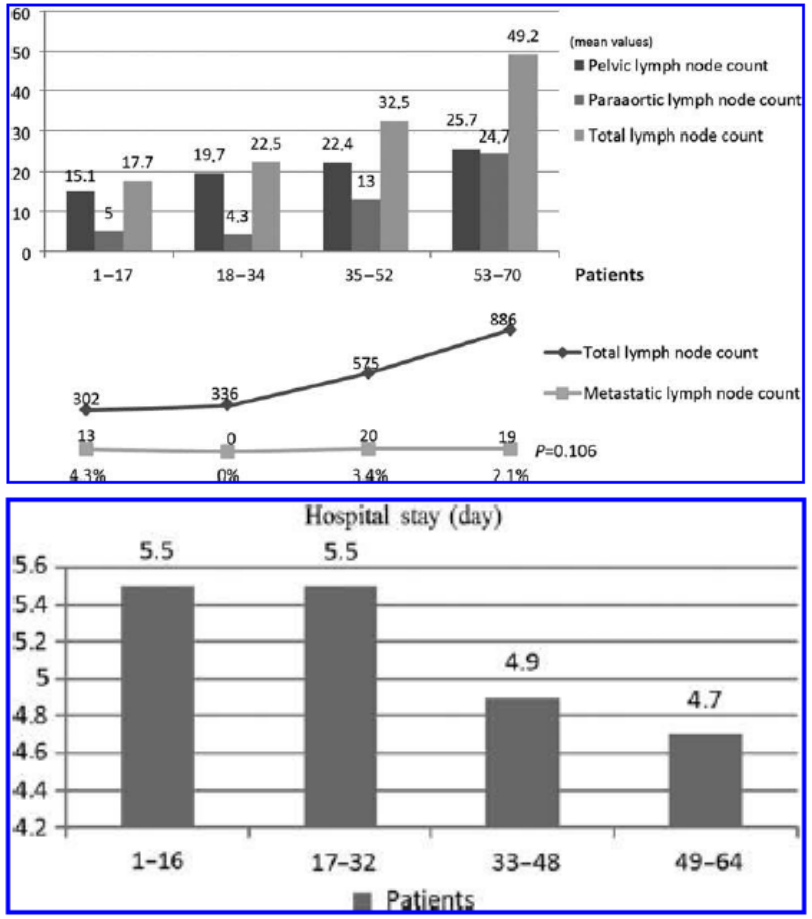
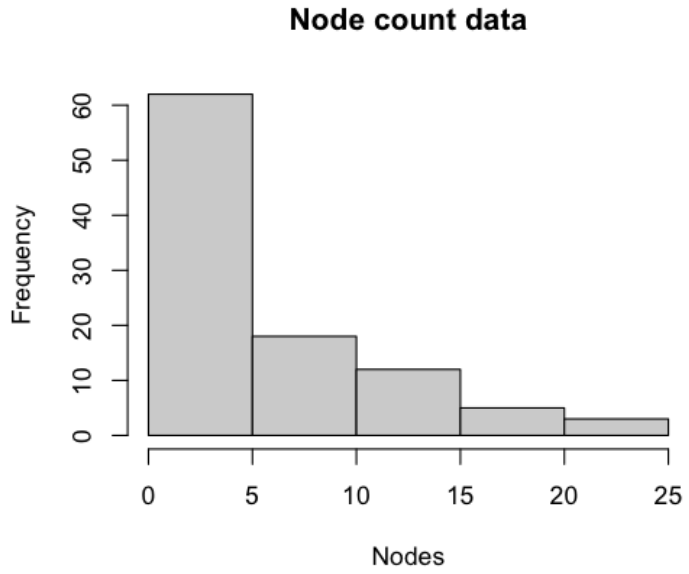


FIG. 7. Hospital stay when data for patients with upper abdominal surgeries were excluded. The slight decrease was not statistically significant ($p = 0.773$).

Kiran *et al*. 2015

- Frequently encountered in medicine.
- Count data as a clinical biomarker, *e.g.*, tumour metases, inflammation in joints, lung noduli, *etc*.
- Count data in preclinical experiments, *e.g.*, cell count, radioactive particles per time.
- In healthcare logistics, *e.g.*, hospital stay in days, number of patients or procedures given a set of conditions (effectors).

Poisson regression

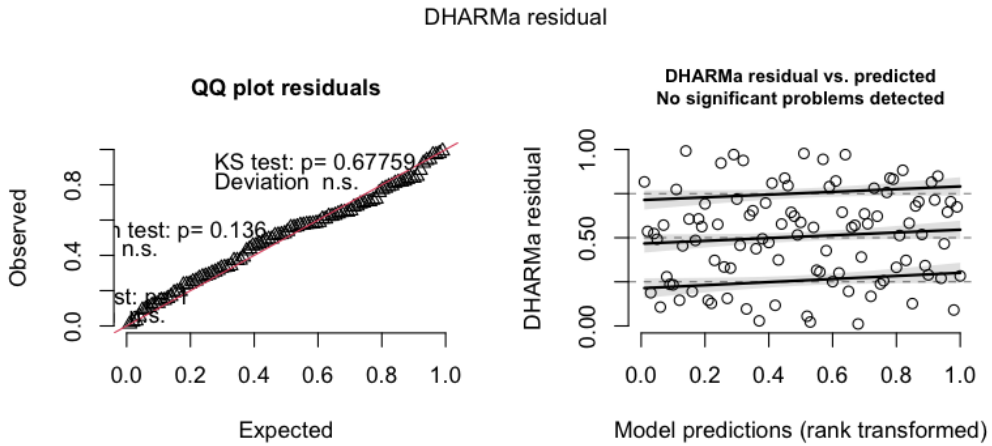


$$y_i \sim Poisson\big(\exp(X_i\beta)\big)$$

Distribution		*Canonical Link:* $\theta = g(\mu)$	*Inverse Link:* $\mu = g^{-1}(\theta)$
Poisson		$\log(\mu)$	$\exp(\theta)$
Binomial	*logit link:*	$\log\left(\frac{\mu}{1-\mu}\right)$	$\frac{\exp(\theta)}{1+\exp(\theta)}$
	probit link:	$\Phi^{-1}(\mu)$	$\Phi(\theta)$
	cloglog link:	$\log(-\log(1-\mu))$	$1-\exp(-\exp(\theta))$
Normal		μ	θ
Gamma		$-\frac{1}{\mu}$	$-\frac{1}{\theta}$
Negative binomial		$\log(1-\mu)$	$1-\exp(\theta)$

- Frequently encountered in medical research.
- Count data as a clinical biomarker, *e.g.*, tumour metases, joint inflammation count, *etc*.
- Count data in preclinical experiments, *e.g.*, cell count, radiation particles.
- In healthcare logistics, *e.g.*, hospital stay in days, number of patients or procedures given a set of conditions (effectors).

Example of Poisson regression in R




```
# DHARMa can be used to produce diagnostic plots of
# most GLMs
# The DHARMa vignette:
# https://cran.r-project.org/web/packages/DHARMa/vignettes/DHARMa.html
library(DHARMa)

sim_res <- simulateResiduals(m1_pois)
plot(res.sim)
```

$$Nodes \sim \beta_0 + \beta_1 Treatment + \beta_2 Biomarker$$

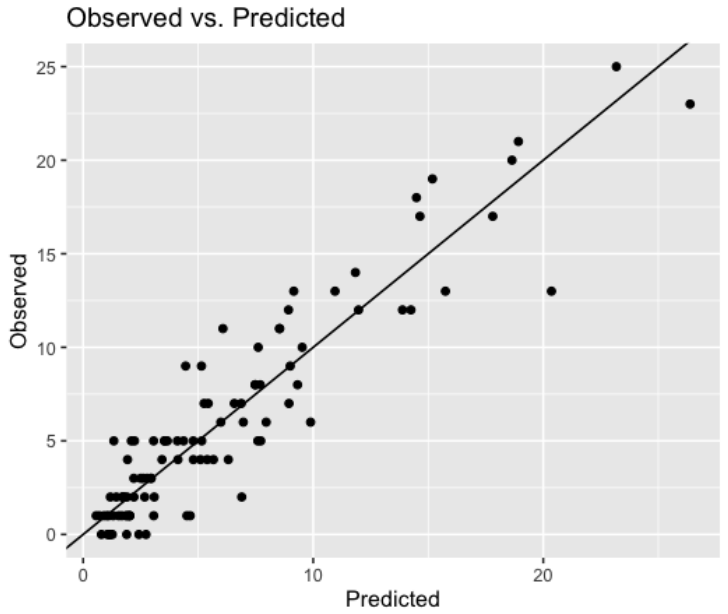
```
> m1_pois <- glm(y ~ x1 + x2, family = "poisson", data = pois_dat)
> summary(m1_pois)
Call:
glm(formula = y ~ x1 + x2, family = "poisson", data = pois_dat)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.02224    0.08455  12.091  < 2e-16 ***
x1           0.50322    0.08638   5.826 5.69e-09 ***
x2           0.74975    0.04078  18.384  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 502.414  on 99  degrees of freedom
Residual deviance:  93.876  on 97  degrees of freedom
AIC: 408.15

Number of Fisher Scoring iterations: 5
```



Overdispersion

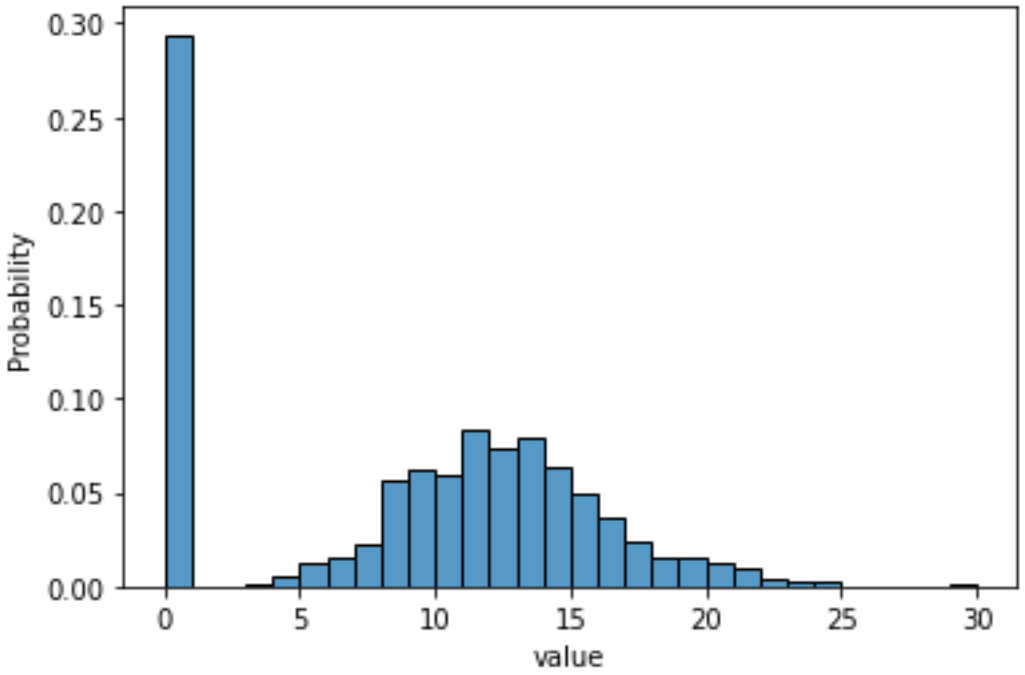
```
>testDispersion(m1_pois)
DHARMa nonparametric dispersion test
via sd of residuals fitted vs.
simulated

data:  simulationOutput
dispersion = 0.73694, p-value =
0.136
alternative hypothesis: two.sided
```

- The Poisson distribution is specified using one parameter, λ, that describes the mean and variance.
- In Poisson regression this can lead to underestimation of the variability in the data.
- We can test for dispersion using a few different methods:
  $Dispersion = Deviance / Degrees\ of\ Freedom$
  $Dispersion = Pearson's\ x^2\ statistic / Degrees\ of\ Freedom$
  R function: `library(DHARMa), testDispersion`.
- Dispersion over 1 indicates overdispersion.
- If overdispersed, try negative binomial regression.
  For example, `library(MASS)`

```
library(MASS)
m2 <- glm.nb(y ~ x1 + x2, data = pois_dat)
```

Zero inflation



- A distribution with frequent zero-valued observations.
- Both Poisson and negative binomial models can handle some zero inflation.
- Switching to a negative binomial model might suffice.
- Use mixture models, *e.g.* Poission and zero-value distribution. Available in for example **`library(pscl)`**

```
library(pscl)
m3_pois <- zeroinfl(y ~ x1 + x2, data =
pois_dat)
```

L3-4 Ordered categorical regression

Ordered categorical data

Table 1. Ordinal variables commonly used in clinical and experimental studies with their quantitative alternatives for data collection.

Variable	Ordinal categories	Quantitative alternative
Functional class	I, II, III, IV	N.A.
Educational level	Illiterate, primary, secondary, higher education	Full years in education
American Society of Anesthesiologists anesthetic risk	I, II, III, IV, V	N.A.
Age group	Children, adolescents, adults, seniors	Age in years
Cancer staging	I, II, III, IV	N.A.
Body composition	Underweight, healthy weight, overweight, obese, morbidly obese	Body mass index
Pallor	0, 1+, 2+, 3+, 4+	Hematocrit
Pulse amplitude	0, 1+, 2+, 3+, 4+	Plethysmography
Histopathological grading	0, 1+, 2+, 3+, 4+	Percentage of cells
Satisfaction	Very unsatisfied, fairly unsatisfied, neutral, satisfied, very satisfied	N.A.
Economic status	Classes A, B, C, D	Family income

N.A. = none available.

1	2	3	4	5
Never	Rarely	Sometimes	Often	Always
Completely disagree	Disagree	Neutral	Agree	Completely agree

Likert scale three, five, seven point scales, *etc*.

- More than two levels of responses.
- Loss of power compared to continuous data, hence can be considered a design aspect *e.g.*, BMI vs underweight, normal, overweight, obese, morbidly obsese.
- Grading of disease severity, risk, conditions, quality. For example, cancer staging or performance status.
- Widely used in diagnostic instruments in psychiatry. For example, frequency of feelings and thoughts, agreement with statements, *etc*.
- Likert data will at occasions be treated as interval data. Opinions differ on the accuracy of this approach.

Miot 2020; Sullivan & Anthony 2013

Ordinal logistic regression

Ordered categorical model with K categories:

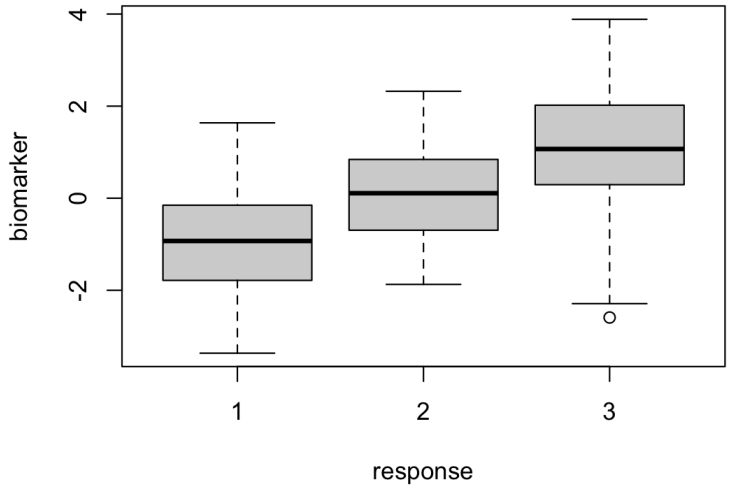
$$y_i = \begin{cases} 1 & if\ z_i < 0 \\ 2\ \ if\ z_i \in (0, c_2) \\ 3\ if\ z_i \in (c_2, c_3) \\ \dots \\ K-1\ if\ \ \in (c_{K-2}, c_{K-1}) \\ K\ if >\ c_{K-1} \end{cases}$$

$$z_i = X_i\beta + \epsilon_i$$

- More than two levels in response.
- Predict *k - 1* log odds, *k* is number of categories.
- Fit one logistic regression to each category, all with the same shape: level 0-1, level 1-2, level 2-3, etc.
- Assumes all curves have the same shape, proportional odds assumption or parallel regression assumption.
- Cut-points (*c*, intercepts) will differ, these are related to thresholds of the response levels. Z is the linear predictor. These will be estimated simultaneously as the coefficients using maximum likelihood estimation.
- The scale σ of the error ϵ corresponds to the fuzziness of the cut-off points.
- We can test the proportional odds assumption using for example Brant's test:

```
library(brant)
brant(model)
```

Ordinal logistic regression




```
Call:
polr(formula = response ~ biomarker, data
= ordinal_dat)

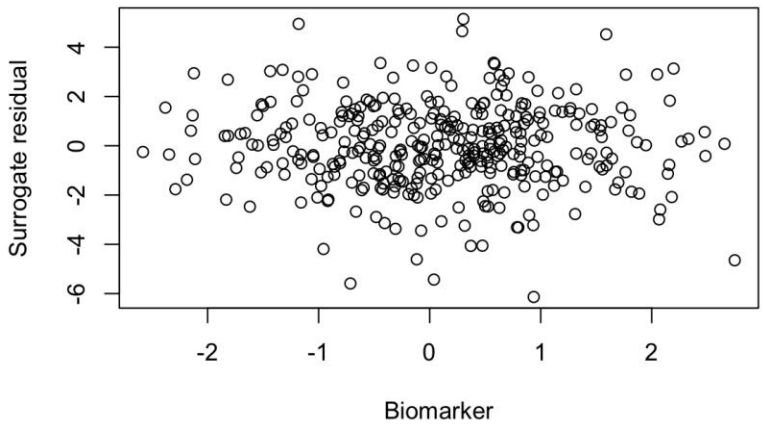
Coefficients:
          Value Std. Error t value
biomarker 1.185     0.1263   9.382

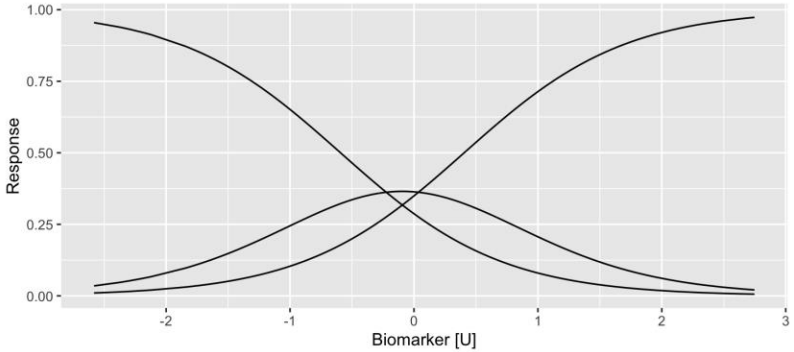
Intercepts:
    Value   Std. Error t value
1|2 -0.8498  0.1558    -5.4548
2|3  0.4925  0.1502     3.2784

Residual Deviance: 445.0645
AIC: 451.0645
```

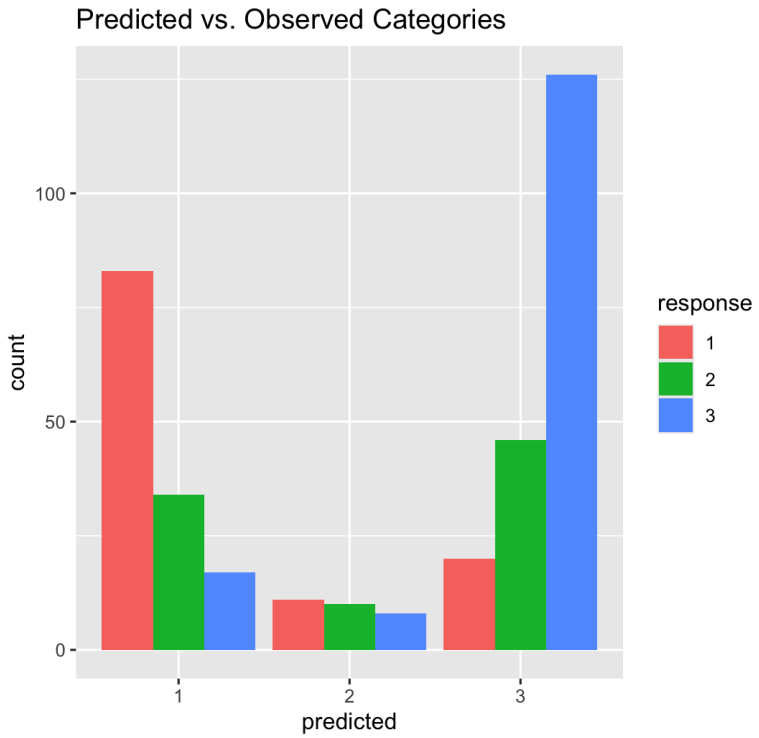
R

```
library(sure)
```





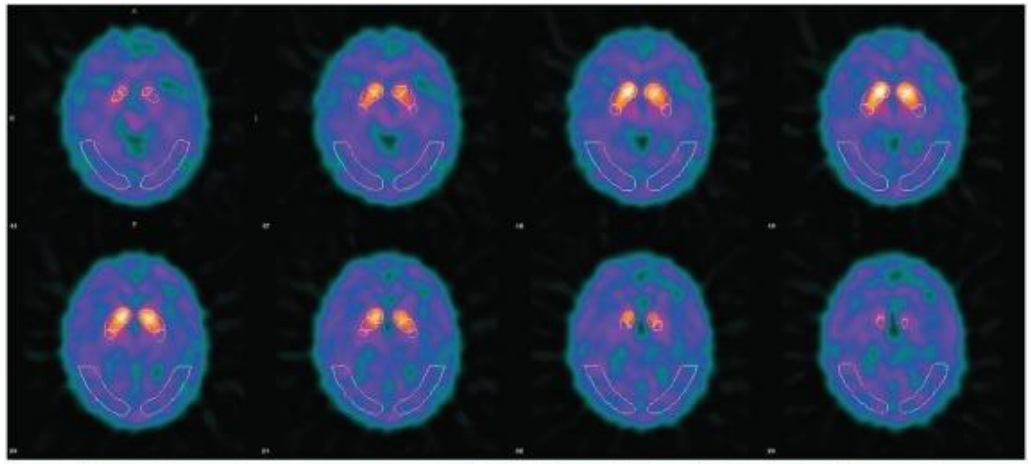
Proportional odds assumption



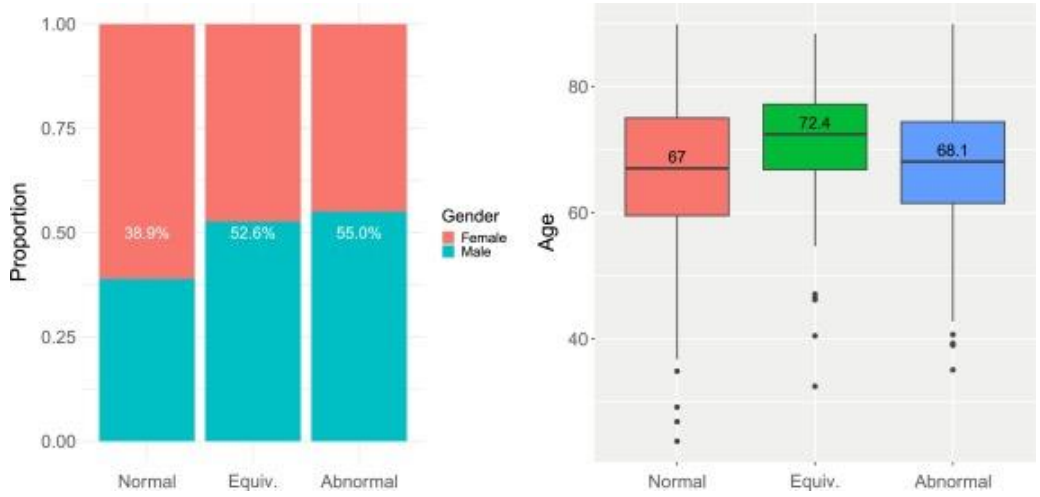
Contingency table:

Actual	Predicted 1	2	3
1	83	11	20
2	34	10	46
3	17	8	126

Modelling unordered categorical data



	Striatum Right	Striatum Left	Anterior Putamen Right	Anterior Putamen Left	Posterior Putamen Right	Posterior Putamen Left	Caudate Right	Caudate Left	Put/Caud Ratio Right	Put/Caud Ratio Left	Putamen Asymmetry	Caudate Asymmetry
Patient uptake	0.82	0.74	0.79	0.77	0.59	0.39	0.95	0.89	0.89	0.88	0.05	0.03

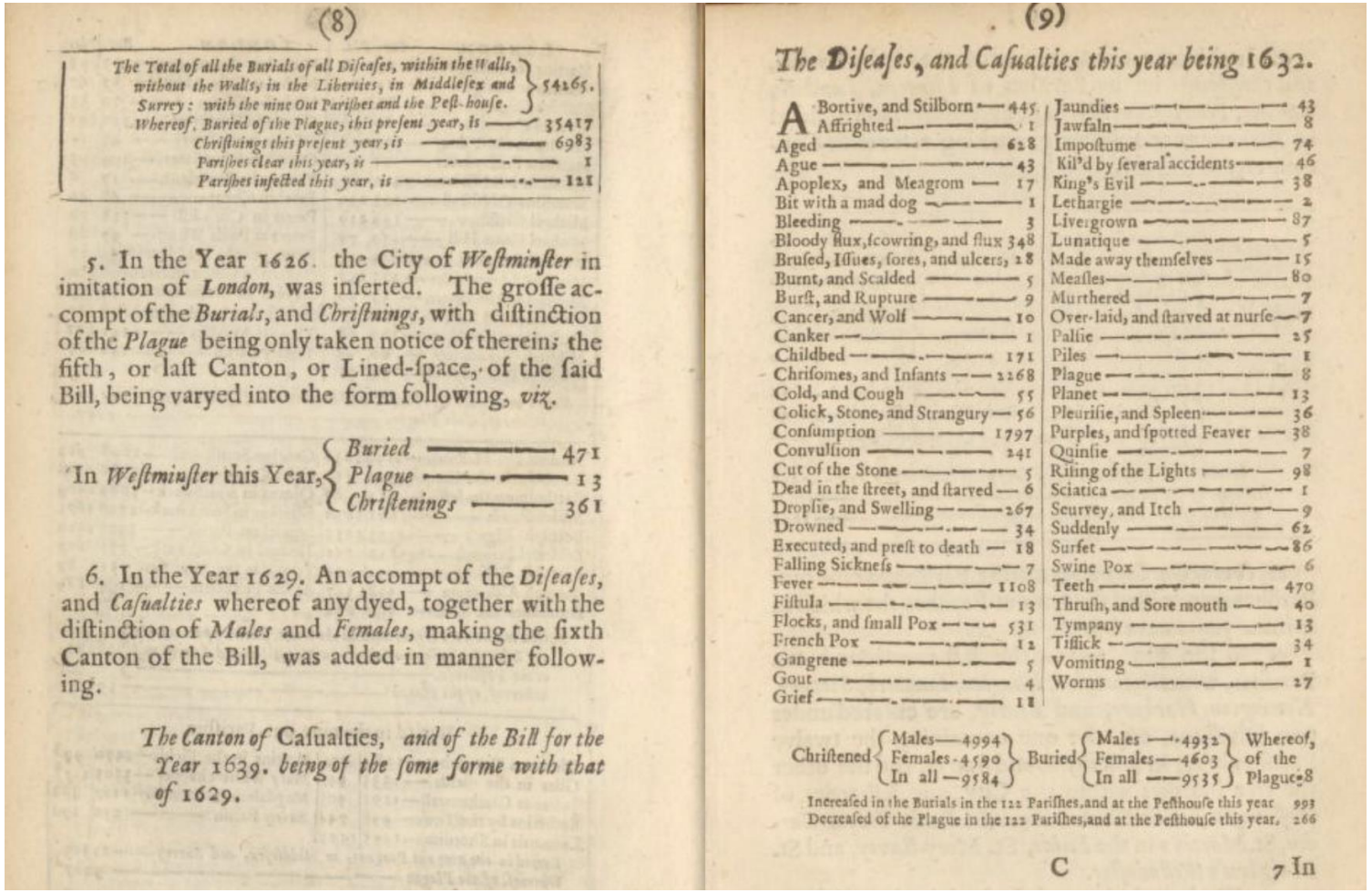


- Modelling of a group of individuals' "choices", without an inherent ordering.
- Alzheimer's disease and cancer, categorisation of asessments (Chu *et al*. 2025): "flexible modelling of interdependence between response categories"
- The binomial probit model for two choices:
  In R: `glm(, family = binomial(probit))`
- Multinomial probit model multiple choices:
  In R: `library(MNP), example(mnp)`
  Vignette: https://cran.r-project.org/web/packages/MNP/vignettes/MNP.pdf

Imai & van Dyk 2005

L3-5 Survival analysis

Survival analysis



(8)

The Total of all the Burials of all Diſeaſes, within the Walls, without the Walls, in the Liberties, in Middleſex and Surrey: with the nine Out Pariſhes and the Peſt-houſe. } 54265.
Whereof, Buried of the Plague, this preſent year, is — 35417
Chriſtnings this preſent year, is — 6983
Pariſhes clear this year, is — 1
Pariſhes infected this year, is — 121

5. In the Year 1626. the City of *Weſtminſter* in imitation of *London*, was inſerted. The groſſe accompt of the *Burials*, and *Chriſtnings*, with diſtinction of the *Plague* being only taken notice of therein; the fifth, or laſt Canton, or Lined-ſpace, of the ſaid Bill, being varyed into the form following, *viz.*

In *Weſtminſter* this Year, { *Buried* — 471 / *Plague* — 13 / *Chriſtenings* — 361 }

6. In the Year 1629. An accompt of the *Diſeaſes*, and *Caſualties* whereof any dyed, together with the diſtinction of *Males* and *Females*, making the ſixth Canton of the Bill, was added in manner following.

The Canton of Caſualties, *and of the Bill for the Year* 1639. *being of the ſame forme with that of* 1629.

(9)

The Diſeaſes, and Caſualties this year being 1632.

Abortive, and Stilborn	445	Jaundies	43
Affrighted	1	Jawfalln	8
Aged	628	Impoſtume	74
Ague	43	Kil'd by ſeveral accidents	46
Apoplex, and Meagrom	17	King's Evil	38
Bit with a mad dog	1	Lethargie	2
Bleeding	3	Livergrown	87
Bloody flux, ſcowring, and flux	348	Lunatique	5
Bruſed, Iſſues, ſores, and ulcers,	28	Made away themſelves	15
Burnt, and Scalded	5	Meaſles	80
Burſt, and Rupture	9	Murthered	7
Cancer, and Wolf	10	Over-laid, and ſtarved at nurſe	7
Canker	1	Palſie	25
Childbed	171	Piles	1
Chriſomes, and Infants	2268	Plague	8
Cold, and Cough	55	Planet	13
Colick, Stone, and Strangury	56	Pleuriſie, and Spleen	36
Conſumption	1797	Purples, and ſpotted Feaver	38
Convulſion	241	Quinſie	7
Cut of the Stone	5	Riſing of the Lights	98
Dead in the ſtreet, and ſtarved	6	Sciatica	1
Dropſie, and Swelling	267	Scurvey, and Itch	9
Drowned	34	Suddenly	62
Executed, and preſt to death	18	Surfet	86
Falling Sickneſs	7	Swine Pox	6
Fever	1108	Teeth	470
Fiſtula	13	Thruſh, and Sore mouth	40
Flocks, and ſmall Pox	531	Tympany	13
French Pox	12	Tiſſick	34
Gangrene	5	Vomiting	1
Gout	4	Worms	27
Grief	11		

Chriſtened { Males — 4994 / Females — 4590 / In all — 9584 } Buried { Males — 4932 / Females — 4603 / In all — 9535 } Whereof, of the Plague 8

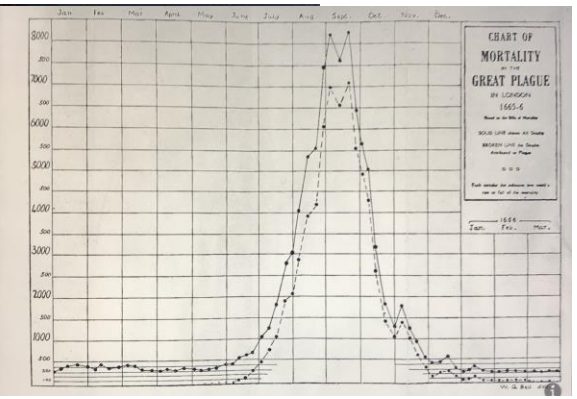
Increaſed in the Burials in the 122 Pariſhes, and at the Peſthouſe this year 993
Decreaſed of the Plague in the 122 Pariſhes, and at the Peſthouſe this year. 266

C 7 In

- Pioneered by Seventeenth Century London Merchant, John Graunt.
- An event as an outcome 0/1.
- Survival analysis is time-to-event analysis accounting for right censoring.
- **Right censoring** survival time is censored at the end of the study, *i.e.* we do not know the survival time of individuals who survived the study.

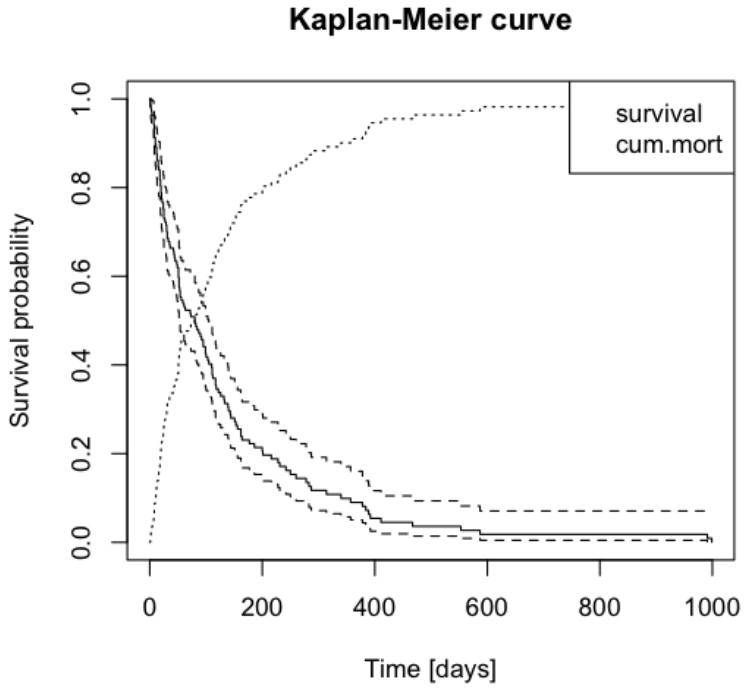






John Graunt first published 1662

Survival analysis



- Participants (or objects) are followed until a terminal event occurs or the end of the study period (right censoring).
- Survival curve, Kaplan-Meier curve (1958), a rough estimate of the cumulative probability of surviving a given time over time. Can be used for group comparisons.
- KM does not provide an effect estimate or related confidence intervals.
- KM does not adjust for confounders.

```
library(survival)
surv_obj <- Surv(data$time, data$event)
surv_km <- survfit(surv_obj ~ 1)
plot(surv_km)
…
survdiff(Surv(…) ~ treatment, data)
```

R

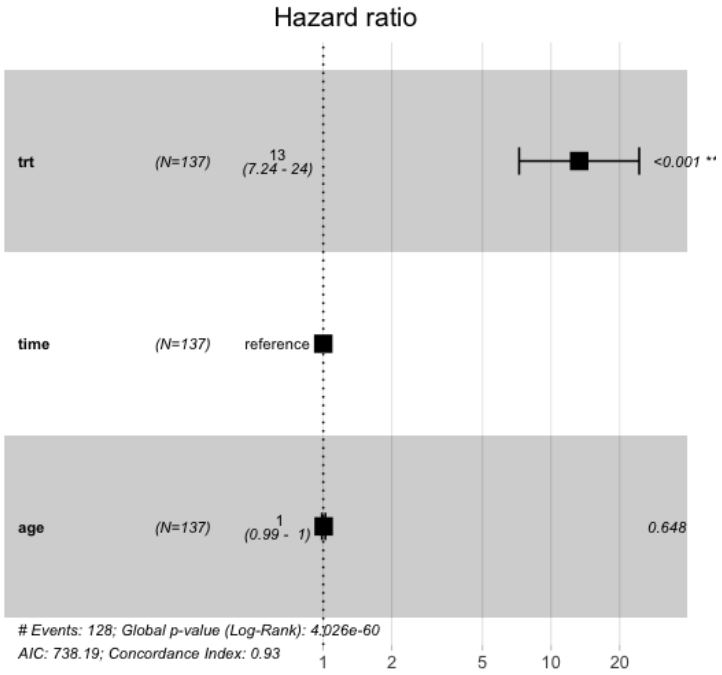
```
> survdiff(vet_obj ~ trt, vet_dat)
Call:
survdiff(formula = vet_obj ~ trt,
data = vet_dat)

       N Observed Expected (O-E)^2/E (O-E)^2/V
trt=1 69       64     64.5   0.00388   0.00823
trt=2 68       64     63.5   0.00394   0.00823

 Chisq= 0  on 1 degrees of freedom, p= 0.9
```

Harrell 2015; Rich *et al*. 2014

Cox proportional hazards regression



- Cox regression (1972) describes the relationship between a time-to-event outcome variable and one or several predictors.
- Dependent variable: incidince rate, independent variables: risk factors or other predictors.
- Semiparametric methods as no assumption on the distribution of survival times, assumes the effect of variables are constant over time (proportionality assumption) and additive over a particular scale.
- $H_0(t)$ is the baseline hazard at time t. The exponential of regression coefficients β_i of x_i give the hazard ratio.

```
library(survival)
library(survminer)
m1_cox <- coxph(Surv(data$time, data$event) ~ trt + age)
cox.zph(m1_cox) # test proportional hazards assumption
ggforest(m1_cox, data)
```

$$H(t) = H_0(t) \cdot \exp(\beta_1 x_1 + \cdots \beta_n x_n)$$

Abd Elhafeez 2021